Chapter 01. Introduction to Investment

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Investment I

Peking University Business School UK Campus

Information about lecture

- Lectures: Monday/Thursday 09:30 – 12:00
- Venue: Lecture Hall
- Class Code: **eb6fvats**
- Prerequisite: business maths

Course Schedule

Time	Content
Lecture 1 Feb 23	Chapter 1. Introduction to Investments • Basics of porfolio mathmatics • Computing return and variance using Excel
Lecture 2 Feb 25	Chapter 2. Capital Allocation to Risky Assets • Stock Risk and Return • Risk aversion and its estimation • Capital allocation between risky and risk-free assets • Passive strategies and the capital market line (CML)
Lecture 3 & 4 Mar 2& Mar 5	Chapter 3. Optimal Risky Portfolio • Optimal risky portfolio • The Markowitz portfolio optimization model • Excel Practice
Lecture 5 Mar 9	Chapter 4. Index Models • Advantages of a single-factor model • Risk decomposition • Systematic vs. firm-specific • Optimal risky portfolio in the single-index model
Lecture 6&7 Mar 12 & Mar 16	Chapter 5. The Capital Asset Pricing Model • Assumption • Capital Market Line • Derivation of CAPM • CAPM and Markowitz • Alpha • Empirical tests of the CAPM

Course Schedule

Time	Content
Lecture 8 Mar 19	Chapter 6. Extensions to CAPM • Low Beta Anomaly and Zero-Beta Model • Non-tradable: Human Capital and Private Business • Liquidity, Size and Value Effect • Consumption-based CAPM
Lecture 9&10 Mar 23 & 26	Chapter 7. Arbitrage Pricing Theory • Factor Model • APT Pricing Equation • Factor-mimicking and factor-tilting • APT Practical Implementation

Lecture 11
 	
Mar 30
	Chapter 8. Performance Evaluation
	• Types of Investment Companies
	• Performance measures:
	  ○ Sharpe ratio
	  ○ Jensen's alpha
	  ○ Treynor's measure
	  ○ Information ratio
	Group Excel Project

Lecture 12 &13 April 2 and 8	Chapter 9. Market Efficiency and Behavioral Finance • Efficient Market Hypothesis (EMH) • Market Efficiency and Return Predictability • Market Efficiency and Behavioral Finance • Limits to Arbitrage
Lecture 14 April 9	Chapter 10. Option basics Revision

Textbook

Bodie, Z., Kane, A., & Marcus, A. (2017). Investments. 11th Edition. McGraw-Hill/Irwin. ISBN-13: 9781526821386 (link). (BKM henceforth)

Assessment

Assessment Task	Weighting
Class Participation	10%
Group Project	20%
Final Exam	70%
Total	100%

Class Participation:
You will be asked to solve problems in the class and be graded according to your effort.
Simply submit your answers to Google Drive and my TA will assess it. Please name the file using **student ID**

Plan (subject to changes):
Exercise of portfolio math (Chapter 01)
Markovitz MV analysis (Chapter 03)
Building optimal portfolio using index models (Chapter 04)

Assessment

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Class Participation	10%
Group Project	20%
Final Exam	70%
Total	100%

Group Project (in the class):

The group project will be conducted in the class on *March 30th*.
The members of group will be announced by the instructor.
<u>Duration: 75 minutes</u>
You will be asked to use Excel to perform optimization and portfolio management using the techniques covered in the class. (for example, building a size portfolio, perform mean-variance analysis)

Assessment

Assessment Task	Weighting
Class Participation	10%
Group Project	20%
Final Exam	70%
Total	100%

Final Exam:

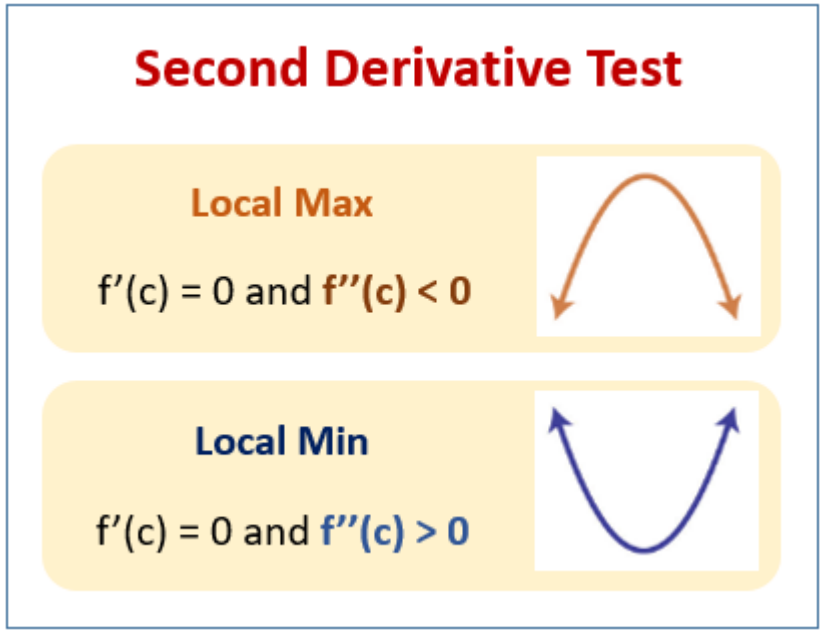
Final exam is a half-open book exam. You are allowed to bring one-page double-sided A4 help note.
No formulas will be provided.

Chapter Outline - Portfolio Mathematics

- Differentiation
- Lagrange Optimization
- Matrix Algebra in Portfolio Management

Differentiation

- Differentiation is a fundamental calculus tool used in optimization to determine the maximum (or minimum) points of a function.
  1. Find the first derivative, set it equal to zero and identify the critical numbers.
  2. Plug the critical numbers into the second derivative function to determine the local concavity of the function to see if its concave or convex.
     - If it's concave - it's a relative maximum.
     - If it’s convex, it's a relative minimum



Differentiation

- Product rule: $[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$

- Ratio rule: $\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$
  - The derivatives of $\frac{e^x}{x}$ is $\frac{e^x x - e^x}{x^2}$
- Chain rule: $[f(g(x))]' = f'\big(g(x)\big)g'(x)$
  - The derivatives of $(3x^2 + 1)^2$ is $2(3x^2 + 1) \times 6x$

Steps of Lagrange Optimization

- Identify Objective and Constraint: Identify the function $f(x, y)$ to be maximized or minimized and the constraint equation $g(x, y) = k$.
- Rearrange Constraint: Rewrite the constraint as $g(x, y) - k = 0$
- Define the Lagrangian: Introduce the Lagrange multiplier λ and define the Lagrangian function:
  - $L(x, y, \lambda) = f(x, y) - \lambda\,(g(x, y) - k)$
  - Calculate Partial Derivatives: Calculate the partial derivatives of L with respect to each variable (x,y, λ), and set them equal to zero:
  - $\frac{\partial L}{\partial x} = 0$
  - $\frac{\partial L}{\partial y} = 0$
  - $\frac{\partial L}{\partial \lambda} = 0$ (This reproduces the constraint $g(x, y) - k = 0$)
- Solve the System

Example

- Assume there are three commodities with amounts x1, x2, and x3, and prices p_1, p_2, and p_3. Assume the total value is fixed, p1x1 + p2x2 + p3x3 = w, where w > 0 is a fixed positive constant.
- Assume the utility is given by U = x1x2x3.
- The maximum of U on the commodity bundles given by the wealth constraint satisfying the equations.

Answer

- $L(x_1, x_2, x_3, \lambda) = x_1 x_2 x_3 - \lambda(p_1 x_1 + p_2 x_2 + p_3 x_3 - w)$
- Partial derivative with respect to x_1, x_2, x_3 is
  - $x_2 x_3 - \lambda p_1$
  - $x_1 x_3 - \lambda p_2$
  - $x_2 x_3 - \lambda p_3$
- Partial derivative with respect to λ is
  - $(p_1 x_1 + p_2 x_2 + p_3 x_3 - w) = 0 \qquad (1)$

By substitution, (1) can be written as

- $3\frac{x_1 x_2 x_3}{\lambda} = w$

- Then, $\frac{x_1 x_2 x_3}{\lambda} = w/3$, and $\frac{x_1 x_2 x_3}{\lambda} = p_1 x_1 = w/3$
- Then $x_1 = w/3p_1$, and $x_2 = w/3p_2$, and $x_3 = w/3p_3$

Standard Notation

- w_i is the weight on Asset i
- R_i is the return on Asset i
  - Sometimes we are interested in expected return: E_i is the expected return on Asset i ($E_i \equiv E(R_i)$)
- σ_i is the standard deviation on Asset i
- ρ_{ij} is the correlation on Asset $i\ and\ j$
- σ_{ij} is the covariance on Asset $i\ and\ j$, $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$

Portfolio Return and Risk

- Portfolio weight for asset i: w_i

$$= \frac{\$\ amount\ in\ asset\ i}{total\ \$\ amount\ in\ the\ portfolio}$$

- By definition: $w_1 + w_2 + \ldots + w_N = 1$
- Long asset i: $w_i > 0$
- Short asset i: $w_i < 0$
- Example (you have \$1000 to invest):
  - Invest \$800 in IBM and \$200 in GE ($w_{GE} = 0.2, w_{IBM} = 0.8$)
  - Short \$1000 of IBM and go long (invest) \$2,000 in GE ($w_{GE} = 2, w_{IBM} = -1$)
- Portfolio returns are linear in the portfolio weights:
- $R_p = w_1 R_1 + w_2 R_2 + \cdots + w_n R_n$
  - Note that, w and R_p can sometimes represent **\$ investment and \$ payoff.**

Example of three assets: Return and Risk

Expected return for three assets:

- $w_1E(R_1) + w_2E(R_2) + w_3E(R_3)$

Portfolio variance for three assets:

- $w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + w_3^2\sigma_3^2 + 2w_1w_2\sigma_{1,2} + 2w_1w_3\sigma_{1,3} + 2w_2w_3\sigma_{2,3}$

However, how about 4, 5 and many assets……

That is why we use matrix to store information of portfolio

Matrix Algebra - Preliminaries

- A matrix is a way of organizing information.
- In portfolio management, we employ the variance-covariance matrix of assets.
- We also write the weight, the expected return in each of the asset in a matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

- A matrix is of order (or dimension) *m* by *n* (also denoted as (*m x n*)).
- All elements can be identified by a typical element a_{ij}, where *i*=1,2,…,*m* denotes rows and *j*=1,2,…,*n* denotes columns.

Transpose

For example, if a (2 x 3) matrix is defined by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

Then the transpose of A, denoted by A', is now (3 x 2)

$$A' = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}$$

A (1x3) matrix is defined as *W*=[0.1 0.3 0.4], then the transpose of *W* is *W'*, which is $\begin{bmatrix} 0.1 \\ 0.3 \\ 0.4 \end{bmatrix}$

A (2x2) matrix is defined as *W*= $\begin{bmatrix} 1 & 3 \\ 4 & 7 \end{bmatrix}$, then the transpose of *W* is *W'*, which is $\begin{bmatrix} 1 & 4 \\ 3 & 7 \end{bmatrix}$

- For a scalar a, $a = a'$
  - A scalar is 1x1.
- If C, D are invertible, then $(CD)' = D'C'$

Identity, Symmetric, and Diagonal matrix

Symmetric matrix	$A = \begin{bmatrix} 4 & 5 & -3 \\ 5 & 7 & 2 \\ -3 & 2 & 10 \end{bmatrix}$	Equal to its transpose, that is, *A=A'* It has the same number of rows as it has columns
Diagonal matrix	$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 10 \end{bmatrix}$	It can also be denoted as *diag(4, 7, 10)* , elsewhere other than the diagonal are zeros
Identity matrix	$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	It can also be denoted as *diag(1, 1, 1)*

Addition and Subtraction

$$A = \begin{bmatrix} 4 & 5 & -3 \\ 5 & 7 & 2 \\ -3 & 2 & 10 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

What is C if C=A + B?

$$C = \begin{bmatrix} 8 & 5 & -3 \\ 5 & 14 & 2 \\ -3 & 2 & 20 \end{bmatrix}$$

What is C if C=A – B?

$$C = \begin{bmatrix} 0 & 5 & -3 \\ 5 & 0 & 2 \\ -3 & 2 & 0 \end{bmatrix}$$

Multiplication

C = A×B, if A is of dimension ($m \times p$) and B of dimension ($p \times n$), then the product, C, will be of order ($m \times n$) whose ijth element is defined as $c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}$

$$A = \begin{bmatrix} 4 & 5 & -3 \\ 5 & 7 & 2 \\ -3 & 2 & 10 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

$$c_{11} = 4 \times 4 + 5 \times 0 + (-3) \times 0 = 16$$

$$c_{12} = 4 \times 4 + 5 \times 7 + (-3) \times 0 = 35$$

What is C if C=A ×B?

$$C = \begin{bmatrix} 16 & 35 & -30 \\ 20 & 49 & 20 \\ -12 & 40 & 100 \end{bmatrix}$$

Note that the **number of columns of A = the number of rows of B**

Multiplication

- Note that $AB \neq BA$
- That is multiplying on the left side is different from multiplying on the right side

Matrix Inverse

- Maximum number of linearly independent columns in a $m \times n$ matrix A is called rank.
  - When the rank of A is $\min(m, n)$, then we say A has full rank.
- For every square matrix A with full rank, there is a matrix B such as $AB = BA = I$
- Matrix B is unique, it is called inverse to matrix A, denoted as A^{-1}
- If C, D are invertible, then $(CD)^{-1} = D^{-1}C^{-1}$

Inverse of a Square Matrix

The inverse of A is denoted by A^{-1} where $AA^{-1} = A^{-1}A = I$ (I indicates identify matrix)
We can use Linear Row Reduction to find the inverse matrix
But for the **(2 x 2)** case, one can apply the following formula

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The inverse of A, denoted as A^{-1}, is

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Portfolio Matrix

- Denote the weighting vector as $\mathrm{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$
- Denote the expected return vector as $\mathrm{E} = \begin{bmatrix} E(r_1) \\ E(r_2) \end{bmatrix}$
- Denote the variance-covariance matrix as $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$

Portfolio Matrix

- Portfolio return $\boldsymbol{\mu} = \mathbf{w}'\mathbf{E}$
  - Note that sometimes w and μ can represent dollar investment and dollar payoff, respectively.
- Portfolio variance $\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}$
- $\boldsymbol{\Sigma}\mathbf{W}$ is a nx1 vector.
  - *i*th element of this vector represents covariance of asset i and market index
- Portfolio constraint $\mathrm{w}'\mathbf{1} = 1$ (bolded $\mathbf{1}$ represents a vector of 1, e.g., $\mathbf{1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$)

Proof (10 mins)

Please prove that

- $\mathbf{w}'\mathbf{E} = w_1 E(R_1) + w_2 E(R_2)$
- $\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w} = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1 w_2 \sigma_{1,2}$
- The 1st element of ΣW is the covariance between asset 1 and market index (= $w_1 r_1 + w_2 r_2$, a portfolio is all assets)
  - $cov(r_1, r_m) = cov(r_1, w_1 r_1 + w_2 r_2) = w_1 Cov(r_1, r_1) + w_2 Cov(r_1, r_2) = w_1\sigma_1^2 + w_2\sigma_{1,2}$
  - $\Sigma W = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{1,2} & \sigma_2^2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, the 1st element of which is $w_1\sigma_1^2 + w_2\sigma_{1,2}$

Example: Two - Assets

- Suppose the market consists of two stocks (A and B respectively). You allocate 20% of your funds to Stock A and 80% of your funds to stock B.
- Hence, the weighting matrix for the investment portfolio can be written as
- $W = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$
- The expected return for stock A is 0.25 and the return for stock B is 0.3. The expected return matrix for the investment portfolio can be written as
- $E = \begin{bmatrix} 0.25 \\ 0.3 \end{bmatrix}$
- The variance of stock A (the variance is the squared term of volatility) is 0.5, the variance of stock B is 0.3. The covariance between them is 0.1. The variance-covariance matrix for the investment portfolio is written as
- $\Sigma = \begin{bmatrix} 0.5 & 0.1 \\ 0.1 & 0.3 \end{bmatrix}$

Questions: What is $W'E$, Σ^{-1}, $W'\Sigma W$,

Solution

- $W'E = 0.2\text{x}0.25+0.3\text{x}0.8 = 0.29$

Expected return of your portfolio

- What is Σ^{-1}?

$$\Sigma^{-1} = \frac{1}{0.14}\begin{bmatrix} 0.3 & -0.1 \\ -0.1 & 0.5 \end{bmatrix} = \begin{bmatrix} 2.14 & -0.714 \\ -0.714 & 3.57 \end{bmatrix}$$

- What is $W'\,\Sigma\,W$?

$$W'\,\Sigma = [0.18\ 0.26]$$

$$W'\,\Sigma\,W = [0.18\ 0.26]\begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} = 0.244$$

Variance of your portfolio;
$\sqrt{0.244} = 0.494$
is the risk/volatility of the fund

Portfolio Matrix I

- P1 is an investment of 70% in stock S1 and 30% in stock S2
- P2 is an investment of 10% in stock S1 and 90% in stock S2
- P3 is an investment of 0% in stock S1 and 100% in stock S2
- If S1 returns 15% and S2 returns 3%, what is the matrix form that calculates the payoff of P1, P2 and P3.
- $\mathrm{w} = \begin{bmatrix} 0.7 & 0.1 & 0 \\ 0.3 & 0.9 & 1 \end{bmatrix} \quad \mathrm{E} = \begin{bmatrix} 15\% \\ 3\% \end{bmatrix}$
- $\mu = \mathrm{w}'\mathrm{E} = \begin{bmatrix} 1.14 \\ 0.42 \\ 0.30 \end{bmatrix}$

Portfolio Matrix II

- If S1 returns 15% and S2 returns 3% in good state, but **10% and 1% in bad state**
- If the portfolio payoff for the three portfolio (denoted as P1, P2, and P3) in good state and bad state is (column vector represents state and row vector represents payoff for each portfolio)

$$\begin{bmatrix} 1.14 & 0.73 \\ 0.42 & 0.19 \\ 0.30 & 0.10 \end{bmatrix}$$

- Question: What is the percentage investment in S1 and S2 for each portfolio?

Portfolio Matrix II

- $w'E = \begin{bmatrix} 1.14 & 0.73 \\ 0.42 & 0.19 \\ 0.30 & 0.10 \end{bmatrix}$ and $E = \begin{bmatrix} 0.15 & 0.1 \\ 0.03 & 0.01 \end{bmatrix}$

Multiple by E^{-1} on the right

- $w' = \begin{bmatrix} 1.14 & 0.73 \\ 0.42 & 0.19 \\ 0.30 & 0.10 \end{bmatrix} E^{-1} = \begin{bmatrix} 70\% & 30\% \\ 10\% & 90\% \\ 0\% & 100\% \end{bmatrix}$

Exercise

Download the exercise: Exercise of Chapter 01

Submit your answer to Google Classroom

Exercise: Maximize Utility Function

Denote the following utility function as

$$U = E(r) - 0.5A\sigma^2$$

- U = Utility
- $E(r)$ = Expected return on the asset or portfolio, which is equal to
  - $r_f + y(E(r_p) - r_f)$
- A = Coefficient of risk aversion
- σ^2 = Variance of returns, which is equal to
  - $y^2\sigma_p^2$
- 0.5 = A scaling factor

- Choose y that maximized the utility function given r_f, $E(r_p)$, σ_p

Exercise: Optimizing the Cobb-Douglas function

A company has determined that its production level is given by the Cobb-Douglas function $F(L, K) = 2.5L^{0.45}K^{0.55}$

where L represents the total number of labor hours in year and K represents the total capital input for the company. Suppose 1 unit of labor costs \$40 and 1 unit of capital costs \$50.

Use the method of Lagrange multipliers to find the maximum value of $F(L, K)$ subject to a budgetary constraint of \$500,000 per year.

Exercise

- P1 is an investment of $700 in stock S1 and $300 in stock S2
- P2 is an investment of $100 in stock S1 and $900 in stock S2
- If the portfolio payoff for P1 is $200 and the portfolio payoff for P2 is $20, use matrix algebra to calculate the stock returns for S1 and S2

Exercise: Portfolio Matrix

- You plan to invest 30% and 70% of your wealth in R1 and R2. The expected return for R1 and R2 is 8% and 10%, respectively. The correlation between R1 and R2 is 0.5. The standard deviation for R1 and R2 is 10% and 20%, respectively.
- Use matrix algebra to calculate the portfolio return $E(r_p)$ and standard deviation σ_p.
- Denote the utility function $U = E(r_p) - 0.5\,\sigma_p^2$, what is the portfolio weight for R1 and R2 that maximizes the utility function.
- Now assume that the correlation between R1 and R2 is 1, denote the utility function $U = E(r_p) - 0.5\,\sigma_p^2$, what is the portfolio weight for R1 and R2 that maximizes the utility function.

Matrix Calculation with Excel

	A	B	C
1	w1	w2	w3

	E	F	G
1	E1	E2	E3
2	σ_1^2	$\sigma_{1,2}$	$\sigma_{1,3}$
3	$\sigma_{1,2}$	σ_2^2	$\sigma_{2,3}$
4	$\sigma_{1,3}$	$\sigma_{2,3}$	σ_3^2

Excel exercise: Excel Exercise of Chapter 01

Portfolio return

MMULT (A1:C1, TRANSPOSE(E1:G1))

Portfolio variance

MMULT(MMULT(A1:C1,E2:G4),TRANSPOSE(A1:C1))

Example

- You invest 20%, 30% and 50% on R1, R2 and R3. The expected return for R1, R2 and R3 is 8%, 10% and 12%, respectively.
- The variance and covariance for the three assets are
- $Var(R_1) = 0.01,\ Var(R_2) = 0.04,\ Var(R_3) = 0.09, Cov(R_1, R_2) = 0.01, Cov(R_1, R_3) = 0.03, Cov(R_2, R_3) = 0.03$
- What is the portfolio return and variance?
- Use excel calculation.

Example - Answer

- You invest 20%, 30% and 50% on R1, R2 and R3. The expected return for R1, R2 and R3 is 8%, 10% and 12%, respectively.
- The variance and covariance for the three assets are
- $Var(R_1) = 0.01,\ Var(R_2) = 0.04,\ Var(R_3) = 0.09, Cov(R_1, R_2) = 0.01, Cov(R_1, R_3) = 0.03, Cov(R_2, R_3) = 0.03$
- What is the portfolio return and variance?

Portfolio Return	0.106
Portfolio Variance	0.0427